

A masked-digit lower bound for the Gyarmati–Hennecart–Ruzsa sum–difference constant

Logan Kleinwaks

First version: July 24, 2026; Version 1.1: July 26, 2026

Abstract

The Gyarmati–Hennecart–Ruzsa sum–difference constant quantifies how large a difference set $X - Y$ can be relative to the sumset $X + Y$ for arbitrarily large finite sets X, Y of integers satisfying $|X + Y| \leq K|X|$. Zheng’s 2025 lower-bound construction encodes bounded integer vectors as base- $(2B + 1)$ integers, allows every digit in $\{0, 1, \dots, B\}$, and imposes a global cap on the sum of the digits. We generalize this construction by allowing the digits to lie in an arbitrary finite mask $M \subseteq \{0, 1, \dots, B\}$. A standard finite-alphabet multinomial estimate, combined with a carry-free digit argument, reduces the resulting asymptotic lower bound to two finite partition polynomials determined by $M + M$ and, for each $d \in M - M$, by the least possible value of $a + b$ among allowed digits $a, b \in M$ with $a - b = d$. Applying the bound to

$$M = \langle 1518, 1524, 1587, 2024, 2032, 2116 \rangle \cap [0, 17032]$$

gives the rigorously certified inequality

$$C_{3a} > 1.19102809.$$

The mask has 3121 elements, $|M + M| = 18730$, and $|M - M| = 32369$. Its generators form the product grid $\{3, 4\} \times \{506, 508, 529\}$, while the column semigroup is a simple gluing with two nearly equal relation degrees. We analyze the exact and near relations induced by this structure and formulate several features that appear to make numerical-semigroup masks effective for the weighted objective arising in the proof.

1 Introduction and the finite-set principle

Let C_{3a} denote the largest exponent θ for which, for every $K > 1$, there are arbitrarily large finite sets $X, Y \subset \mathbb{Z}$ such that

$$|X + Y| \leq K|X|, \quad |X - Y| \geq c(K)|X + Y|^\theta$$

for some $c(K) > 0$. Gyarmati, Hennecart and Ruzsa proved the following finite-set lower-bound principle [3]: if $U \subset \mathbb{N}_0$ is finite and contains zero, and if

$$|U - U| < 2 \max U + 1,$$

then

$$C_{3a} \geq 1 + \frac{\log(|U - U|/|U + U|)}{\log(2 \max U + 1)}. \quad (1)$$

Recent computational work on this constant includes AlphaEvolve [4], Gerbicz’s large finite construction [5], and Zheng’s large-deviation construction [6]. The *Optimization Constants in Mathematics* repository [2] subsequently listed the finite-certificate bounds 1.1740744 and 1.1835129324

[7, 8]. During preparation of this note, a further open submission to that repository proposed 1.187326127925948 [9]. The bound proved here exceeds all of these values.

Gerbicz introduced bounded base digits into the original construction, and Zheng analyzed the limiting exponential growth of a bounded-digit family. In Zheng's version the allowed digit alphabet is the complete interval $\{0, \dots, B\}$. We replace that interval by a mask M . This can remove many possible sum digits while preserving difference digits d that admit witnesses $a, b \in M$ with $a - b = d$ and small $a + b$, so that they use little of the two digit-sum budgets in the construction. The multinomial counting estimate used below is standard; Proposition 1 is the construction-specific arbitrary-mask consequence.

2 The masked-digit bound

Let $M \subset \mathbb{N}_0$ be finite with $0 \in M$ and $B = \max M \geq 1$. Let

$$Q = 2B + 1, \quad S = M + M, \quad D = M - M.$$

For each $d \in D$, define its minimum witness cost

$$\kappa(d) = \min\{a + b : a, b \in M, a - b = d\}. \quad (2)$$

The function κ is even: $\kappa(-d) = \kappa(d)$. Define

$$P_+(x) = \sum_{s \in S} x^s, \quad P_-(x) = \sum_{d \in D} x^{\kappa(d)}. \quad (3)$$

We use the following standard lower bound for a multinomial coefficient; see, for example, Csiszár [1, Lemmas II.1 and II.2]. If E is a nonempty finite set and nonnegative integers r_e satisfy $\sum_{e \in E} r_e = m$, let $q_e = r_e/m$. Then

$$\log \frac{m!}{\prod_{e \in E} r_e!} \geq -m \sum_{e \in E} q_e \log q_e - \log \binom{m + |E| - 1}{|E| - 1}, \quad (4)$$

where $\log$ is the natural logarithm and $0 \log 0$ is interpreted as zero. For fixed E , the binomial coefficient in (4) is a polynomial in m of degree $|E| - 1$; equivalently,

$$\log \binom{m + |E| - 1}{|E| - 1} = O_E(\log m) = o(m).$$

Proposition 1 (Masked-digit bound). *For every $x \in (0, 1)$,*

$$C_{3a} \geq 1 + \frac{\log P_-(x) - \log P_+(x)}{\log(2B + 1)}. \quad (5)$$

Consequently, one may take the supremum of the right-hand side over $x \in (0, 1)$.

Proof. Fix $x \in (0, 1)$ and define

$$p_d = \frac{x^{\kappa(d)}}{P_-(x)} \quad (d \in D). \quad (6)$$

Since κ is even, these weights satisfy $p_d = p_{-d}$. Let

$$\mu = \sum_{d \in D} p_d \kappa(d).$$

The entropy of p is

$$G(p) = - \sum_{d \in D} p_d \log p_d = \log P_-(x) - \mu \log x.$$

We have $G(p) < \log Q$. Indeed, $|D| \leq Q$. If $|D| < Q$, this follows from $G(p) \leq \log |D|$. If $|D| = Q$, the distribution is not uniform: $p_0 = 1/P_-(x)$, whereas $p_d = x^{\kappa(d)}/P_-(x) < p_0$ for $d \neq 0$.

For each $d \in D$, choose $\alpha_d, \beta_d \in M$ such that

$$\alpha_d - \beta_d = d, \quad \alpha_d + \beta_d = \kappa(d),$$

with $(\alpha_{-d}, \beta_{-d}) = (\beta_d, \alpha_d)$; for $d = 0$, take $(\alpha_0, \beta_0) = (0, 0)$.

Let

$$R = \log \frac{P_-(x)}{P_+(x)}.$$

If $R \leq 0$, then (5) follows from the elementary bound $C_{3a} \geq 1$ (take $Y = \{0\}$ and arbitrarily large finite X). Assume henceforth that $R > 0$, and choose

$$0 < \varepsilon < \min \left\{ \frac{R}{-\log x}, \frac{\log Q - G(p)}{-\log x} \right\}.$$

For each positive $d \in D$, define

$$n_d^{(m)} = n_{-d}^{(m)} = \lfloor mp_d \rfloor, \quad n_0^{(m)} = m - 2 \sum_{\substack{d \in D \\ d > 0}} \lfloor mp_d \rfloor.$$

These are nonnegative integers, their sum is m , and $n_d^{(m)}/m \rightarrow p_d$ for every $d \in D$. Write $q_d^{(m)} = n_d^{(m)}/m$. The number of sequences containing exactly $n_d^{(m)}$ copies of each d is the multinomial coefficient

$$\frac{m!}{\prod_{d \in D} n_d^{(m)}!}.$$

By (4), its logarithm is at least

$$-m \sum_{d \in D} q_d^{(m)} \log q_d^{(m)} - \log \binom{m + |D| - 1}{|D| - 1} = mG(p) + o(m).$$

Indeed, $q_d^{(m)} \rightarrow p_d$ for every d , the function $t \mapsto -t \log t$ is continuous on $[0, 1]$ when its value at zero is taken to be zero, and the final logarithmic term is $o(m)$ because D is fixed. Also, for all sufficiently large m ,

$$\sum_{d \in D} n_d^{(m)} \kappa(d) \leq (\mu + \varepsilon)m.$$

Set

$$L_m = \left\lfloor \frac{(\mu + \varepsilon)m}{2} \right\rfloor$$

and define

$$U_m = \left\{ \sum_{i=0}^{m-1} a_i Q^i : a_i \in M, \sum_{i=0}^{m-1} a_i \leq L_m \right\}. \quad (7)$$

Take any sequence $(d_0, \dots, d_{m-1})$ containing exactly $n_d^{(m)}$ copies of each $d \in D$. Symmetry of the counts gives $\sum_i d_i = 0$. Hence

$$\sum_i \alpha_{d_i} - \sum_i \beta_{d_i} = 0, \quad \sum_i \alpha_{d_i} + \sum_i \beta_{d_i} = \sum_i \kappa(d_i),$$

so each of the first two sums equals one half of the last. Each is an integer not exceeding $(\mu + \varepsilon)m/2$, and hence is at most L_m . Therefore

$$u = \sum_{i=0}^{m-1} \alpha_{d_i} Q^i, \quad v = \sum_{i=0}^{m-1} \beta_{d_i} Q^i$$

belong to U_m , and

$$u - v = \sum_{i=0}^{m-1} d_i Q^i.$$

The map from difference-digit sequences to integers is injective. Indeed, if

$$\sum_i d_i Q^i = \sum_i d'_i Q^i,$$

then, with $\delta_i = d_i - d'_i$, we have $\sum_i \delta_i Q^i = 0$ and $|\delta_i| \leq 2B < Q$. Reduction modulo $Q = 2B + 1$ therefore gives $\delta_0 = 0$; after division by Q , induction gives $\delta_i = 0$ for every i . The multinomial estimate above therefore yields

$$\liminf_{m \rightarrow \infty} \frac{1}{m} \log |U_m - U_m| \geq G(p). \quad (8)$$

Every element of $U_m + U_m$ has a carry-free base- Q digit sequence $(s_0, \dots, s_{m-1}) \in S^m$ with $\sum_i s_i \leq 2L_m$. If N_m is the number of all such sequences, then, because $0 < x < 1$,

$$N_m x^{2L_m} \leq \sum_{(s_i) \in S^m} x^{\sum_i s_i} = P_+(x)^m.$$

Thus

$$\limsup_{m \rightarrow \infty} \frac{1}{m} \log |U_m + U_m| \leq \log P_+(x) - (\mu + \varepsilon) \log x. \quad (9)$$

Combining (8), the identity defining $G(p)$, and (9) gives

$$\liminf_{m \rightarrow \infty} \frac{1}{m} \log \frac{|U_m - U_m|}{|U_m + U_m|} \geq R + \varepsilon \log x > 0. \quad (10)$$

We must also verify the hypothesis of the finite-set principle (1). Every difference in $U_m - U_m$ has a unique signed base- Q digit sequence $(d_i) \in D^m$. If it comes from digit sequences (a_i) and (b_i) in U_m , then

$$\sum_i \kappa(d_i) \leq \sum_i (a_i + b_i) \leq 2L_m.$$

Let T_m count all sequences in D^m satisfying this cost bound. Since $0 < x < 1$,

$$T_m x^{2L_m} \leq \sum_{(d_i) \in D^m} x^{\sum_i \kappa(d_i)} = P_-(x)^m.$$

It follows that

$$\limsup_{m \rightarrow \infty} \frac{1}{m} \log |U_m - U_m| \leq \log P_-(x) - (\mu + \varepsilon) \log x = G(p) - \varepsilon \log x < \log Q, \quad (11)$$

where the final inequality is part of the choice of ε .

Every $u \in U_m$ has digits at most B , so

$$u \leq B \sum_{i=0}^{m-1} Q^i = \frac{Q^m - 1}{2}.$$

On the other hand, $\mu > 0$ because $B \in D$ and $p_B > 0$, so $L_m \geq B$ for all sufficiently large m ; then $BQ^{m-1} \in U_m$. Consequently, if $q_m = 2 \max U_m + 1$, then

$$(Q - 1)Q^{m-1} + 1 \leq q_m \leq Q^m$$

for all sufficiently large m , and hence

$$\lim_{m \rightarrow \infty} \frac{1}{m} \log q_m = \log Q.$$

Together with (11), this proves $|U_m - U_m| < q_m$ for all sufficiently large m , as required in (1).

The numerator in (1) is positive for all sufficiently large m by (10); hence replacing its denominator by the larger quantity $m \log Q$ preserves a lower bound. Applying (1) and taking the lower limit of the resulting right-hand side gives

$$C_{3a} \geq 1 + \frac{R + \varepsilon \log x}{\log Q}.$$

Letting $\varepsilon \downarrow 0$ proves (5). □

Remark 1 (Recovery of Zheng's optimization). *For the complete mask $M = \{0, 1, \dots, B\}$,*

$$P_+(x) = 1 + x + \dots + x^{2B}, \quad P_-(x) = 1 + 2x + 2x^2 + \dots + 2x^B.$$

Maximizing Proposition 1 for $B = 5$ reproduces Zheng's value 1.173077279785... [6]. Thus the proposition is a direct mask generalization of Zheng's limiting construction, although the proof above is self-contained apart from the finite-set principle (1) and the standard multinomial estimate (4).

Remark 2 (The balancing condition at an interior optimum). *Write $x = e^{-\lambda}$ and*

$$F(\lambda) = \log P_-(e^{-\lambda}) - \log P_+(e^{-\lambda}).$$

Give S probability weights proportional to $e^{-\lambda s}$ and give D probability weights proportional to $e^{-\lambda \kappa(d)}$. Direct differentiation gives

$$F'(\lambda) = \mathbb{E}_+[s] - \mathbb{E}_-[\kappa(d)].$$

Consequently, every interior stationary point satisfies

$$\mathbb{E}_+[s] = \mathbb{E}_-[\kappa(d)].$$

This identity is useful when comparing masks: the optimizing parameter balances the typical weighted sum digit against the typical minimum difference-witness cost. It is not needed for the proof, which uses only one fixed value of λ .

3 The certified mask

Let

$$H = \langle 1518, 1524, 1587, 2024, 2032, 2116 \rangle \quad (12)$$

and let

$$M = H \cap [0, 17032].$$

Exact integer calculation gives

$$B = \max M = 17032, \quad Q = 2B + 1 = 34065,$$

and

$$|M| = 3121, \quad |M + M| = 18730, \quad |M - M| = 32369. \quad (13)$$

The conductor of H is 28922. Moreover,

$$[-13066, 13066] \cap \mathbb{Z} \subseteq M - M,$$

and the cheapest witness for difference one has cost

$$\kappa(1) = 14351.$$

The verification package contains the complete mask, complete sum support, and all values $(d, \kappa(d))$ for $d \in M - M$.

At the fixed decimal value

$$\lambda_0 = 0.000321149844434550835903084464712287774157626054669874186964, \quad x_0 = e^{-\lambda_0},$$

the partition sums are approximately

$$P_-(x_0) = 668.3247325490738970446 \dots, \quad P_+(x_0) = 91.03099370518693032667 \dots,$$

so

$$\log \frac{P_-(x_0)}{P_+(x_0)} = 1.99357414328979926593 \dots$$

and

$$1 + \frac{\log(P_-(x_0)/P_+(x_0))}{\log 34065} = 1.19102809750994590261 \dots \quad (14)$$

The accompanying MPFR verifier uses 384-bit arithmetic with directed rounding. It reconstructs all discrete data from the six generators and proves the conservative strict inequality

$$\boxed{C_{3a} > 1.19102809}. \quad (15)$$

No optimization claim is needed for the proof: Proposition 1 holds at the single fixed value λ_0 .

4 Structure of the successful masks

A numerical semigroup is a cofinite additive submonoid of $\mathbb{N}_0$. Numerical semigroups are useful here because their missing elements suppress the low-degree part of P_+ , while additive relations among their generators create further sum collisions. The objective is nevertheless not determined by the Hilbert series or by the three cardinalities in (13). It also depends on the complete weighted minimum-cost function κ .

For a generating vector $G = (g_1, \dots, g_e)$, a useful untruncated analogue of κ is the residual norm

$$\nu_G(d) = \min \left\{ \sum_i (z_i^+ + z_i^-) g_i : z_i^\pm \in \mathbb{N}_0, \sum_i (z_i^+ - z_i^-) g_i = d \right\}.$$

Thus $\nu_G(d)$ is the least total cost of two semigroup elements whose difference is d . If a minimizing pair lies in $[0, B]$, then $\kappa(d) = \nu_G(d)$. Exact relations, with residual zero, help produce sum collisions; near relations with small nonzero residuals help produce many difference digits at low cost. The record combines both kinds.

4.1 The progression to the record

The record emerged from a sequence of increasingly structured numerical-semigroup masks:

semigroup generators	B	resulting value
$\{24, 26, 36, 39\}$	189	1.1893936243 ...
$\{60, 63, 80, 84\}$	538	1.1902074763 ...
$\{312, 315, 336, 416, 420\}$	3084	1.1902381380 ...
$\{1518, 1524, 1587, 2024, 2032, 2116\}$	17032	1.1910280975 ...

The first two generator sets have the forms

$$12\langle 2, 3 \rangle + 13\langle 2, 3 \rangle \quad \text{and} \quad 20\langle 3, 4 \rangle + 21\langle 3, 4 \rangle.$$

In the general four-generator family $S\langle k, k+1 \rangle + (S+1)\langle k, k+1 \rangle$, the identity

$$Sk + (S+1)(k+1) = S(k+1) + (S+1)k + 1$$

produces a unit residual with shared support.

The former five-generator record is an iterated gluing:

$$\langle 312, 315, 336, 416, 420 \rangle = 21\langle 15, 16, 20 \rangle + 104\langle 3, 4 \rangle, \quad \langle 15, 16, 20 \rangle = 5\langle 3, 4 \rangle + 16\mathbb{N}_0.$$

Gluing is standard in the theory of complete-intersection numerical semigroups [10, 11]. Studying this decomposition led to searches parameterized by relations rather than by unrelated generator coordinates.

4.2 A balanced simple gluing in the columns

Write

$$C = \langle 506, 508, 529 \rangle.$$

The column semigroup has the exact simple-gluing description

$$C = 23\langle 22, 23 \rangle + 508\mathbb{N}_0, \quad 508 = 21 \cdot 22 + 2 \cdot 23. \tag{16}$$

More generally, for a simple gluing

$$C_{a,b,r,s} = a\langle r, s \rangle + b\mathbb{N}_0 = \langle ar, as, b \rangle,$$

where $\gcd(r, s) = \gcd(a, b) = 1$ and $b \in \langle r, s \rangle$, the two complete-intersection relation degrees are ars and ab , and its Hilbert series is

$$\sum_{n \in C_{a,b,r,s}} t^n = \frac{(1 - t^{ars})(1 - t^{ab})}{(1 - t^{ar})(1 - t^{as})(1 - t^b)}.$$

For the record parameters

$$(r, s, a, b) = (22, 23, 23, 508),$$

the relation degrees are

$$ars = 11638, \quad ab = 11684,$$

differing by only 46, while the generators

$$ar = 506, \quad b = 508, \quad as = 529$$

are also clustered. The two balances have distinct meanings: $b \approx rs$ aligns the relation degrees, whereas $b \approx ar$ clusters two generators. Satisfying both naturally forces $a \approx s$; in the record, $a = s$ exactly. Formula (16) also gives the column Frobenius number

$$23(22 \cdot 23 - 22 - 23) + 22 \cdot 508 = 21779.$$

4.3 The 2-by-3 generator grid

The record semigroup is generated by the grid

$$\{3, 4\} \times \{506, 508, 529\} = \{1518, 1524, 1587, 2024, 2032, 2116\}. \quad (17)$$

For a column c , the two row generators have the exact relation

$$4(3c) = 3(4c).$$

For the three record columns, these relations occur at degrees

$$6072, \quad 6096, \quad 6348,$$

well before the cutoff $B = 17032$.

The column relation $23 \cdot 506 = 22 \cdot 529$ also lifts to the grid:

$$5(1518) + 2(2024) = 2(1587) + 4(2116) = 11638.$$

The other relation in (16) can be lifted after adding $4 \cdot 529$ to both sides:

$$5(1524) + 2(2032) + 2116 = 7(1518) + 2(1587) = 13800.$$

Thus several exact relation layers are already active below the chosen cutoff, while the conductor of the full six-generator semigroup, 28922, remains substantially larger.

The same grid simultaneously gives cheap near relations. If $c_i < c_j$ are columns, then

$$(3c_i + 4c_j) - (4c_i + 3c_j) = c_j - c_i, \quad (18)$$

and the total cost of the two witnesses is $7(c_i + c_j)$. The column gaps are 2, 21, and 23, so the mask has the explicit bounds

$$\kappa(2) \leq 7098, \quad \kappa(21) \leq 7259, \quad \kappa(23) \leq 7245.$$

Because $\gcd(2, 21) = 1$, the cheap residual directions generate the full integer lattice, although combining them can increase witness cost and is subject to the cutoff. In contrast, the exact certificate gives $\kappa(1) = 14351$. This illustrates why minimizing $\kappa(1)$ alone is not an adequate design criterion: Proposition 1 uses the entire weighted distribution of $\kappa(d)$.

4.4 What appears to matter

Searches around (17) found a narrow interior ridge: consecutive r, s , equality $a = s$, near equality $b \approx rs \approx ar$, and the row pair $\{3, 4\}$ were all favored. Relaxing the relation balance and adding a third binary product factor did not improve the record. These computations are evidence about the explored families, not an optimality proof.

The structural evidence suggests that a strong semigroup mask should have:

1. several exact relations becoming active below the cutoff;
2. several cheap near relations with small coprime residuals;
3. substantial sharing of generators among those relations;
4. clustered relation degrees, so that multiple collision mechanisms become active in the same range; and
5. delayed density, with the cutoff after the useful relation layers but before the conductor.

Product grids and gluings manufacture these features, but neither is logically required by Proposition 1. A natural broader program is therefore to prescribe a small weighted relation lattice first and solve the resulting Diophantine constraints for the generators. This includes low-rank grids with small circuit defects, carefully selected additive gluings, and low-dimensional Kunz-cone faces. Such a relation-template approach is more closely tied to the objective than an unstructured search over generator tuples.

5 Verification and reproducibility

The source code, certificate files, verification scripts, and saved run logs are available at

<https://github.com/kleinwaks/masked-digit-sum-difference-bound>.

The package contains the following independently written checks.

1. `verification/verify_mpfr.cpp` reconstructs M , $M + M$, and $\kappa(d)$ using exact integer arithmetic and certifies (15) using MPFR directed rounding.
2. `verification/verify_python.py` independently reconstructs the exact discrete data using the Python standard library and checks the shipped certificate files and hashes.
3. `verification/verify_gp.gp` independently reconstructs the data in PARI/GP and evaluates the bound at high precision. This is a secondary numerical cross-check; the MPFR program supplies the rigorous floating-point certificate.

Acknowledgements and AI-use disclosure

During this work, the author used several large language models (LLMs). OpenAI’s GPT-5.6 Sol model was used extensively in software development, analysis of search results, drafting and revising this paper, and preparation of the verification package. Google’s Gemini 3.1 Pro Preview was used in software development and analysis of search results; Google’s Gemini 3.5 Flash was used to review the paper; and Anthropic’s Sonnet 5 was used in analysis of search results. The author set the research goals, directed the computational work, selected which suggestions to pursue, ran the search and verification software, reviewed the mathematical arguments and outputs, and takes responsibility for all claims and references.

References

- [1] I. Csiszár, *The method of types*, IEEE Transactions on Information Theory 44 (1998), no. 6, 2505–2523, https://iest2.ie.cuhk.edu.hk/~whyung/post/manuscript/06_SAEF/method%20of%20types.pdf.
- [2] D. Davis, P. Ivanisvili, T. Tao, and contributors, *Optimization Constants in Mathematics*, GitHub repository, 2026, <https://github.com/teorth/optimizationproblems>.
- [3] K. Gyarmati, F. Hennecart, and I. Z. Ruzsa, *Sums and differences of finite sets*, Functiones et Approximatio Commentarii Mathematici 37 (2007), 175–186, <https://gyarmatikati.web.elte.hu/publ/sumdiffv.pdf>.
- [4] B. Georgiev, J. Gómez-Serrano, T. Tao, and A. Z. Wagner, *Mathematical exploration and discovery at scale*, arXiv:2511.02864, 2025, <https://arxiv.org/abs/2511.02864>.
- [5] R. Gerbicz, *Sums and differences of sets (improvement over AlphaEvolve)*, arXiv:2505.16105, 2025, <https://arxiv.org/abs/2505.16105>.
- [6] F. Zheng, *Sums and differences of sets: a further improvement over AlphaEvolve*, arXiv:2506.01896, 2025, <https://arxiv.org/abs/2506.01896>.
- [7] S. Griego, *Base-21 digit construction certificate for C_{3a}* , GitHub pull request #71 to the Optimization Constants repository, 2026, <https://github.com/teorth/optimizationproblems/pull/71>.
- [8] Mosaic Intelligence, *Exact-count certificate for problem 3a*, GitHub pull request #95 to the Optimization Constants repository, 2026, <https://github.com/teorth/optimizationproblems/pull/95>.
- [9] Numaro, *Certified lower bound $C_{3a} \geq 1.187326127925948$* , GitHub pull request #131 to the Optimization Constants repository, 2026, <https://github.com/teorth/optimizationproblems/pull/131>.
- [10] A. Assi and P. A. García-Sánchez, *Constructing the set of complete intersection numerical semigroups with a given Frobenius number*, arXiv:1204.4258, 2013, <https://arxiv.org/abs/1204.4258>.
- [11] J. C. Rosales and P. A. García-Sánchez, *Numerical Semigroups*, Developments in Mathematics, vol. 20, Springer, 2009.